

WARNING!

USE A USB CABLE WITHOUT THE 5 VOLT PIN

**ALWAYS CHECK WITH A MULTIMETER: THE 5 VOLT PIN
MUST NOT CARRY 5 VOLTS AND NO PIN MAY GET MORE
THAN 3.3 VOLTS.**

OTHERWISE THE MISTER CAN BREAK!

OutRunners for two: simple guide

How to link two MiSTer FPGAs (Multisystem2 included) and play together. Version 2.

1. The cable and the extension

You need a **USB 3.0 cable with a Type-A plug on both ends** (male-to-male), connected to the **USER port** of both machines (the blue USB port, which is not a real USB port). In this cable the 5 V wire (**pin 1, called VBUS**) **must be interrupted**: joining the 5 V of two machines can damage them. The other wires stay as they are.

You can add a normal **USB 3.0 male-to-female extension without removing the 5 V pin** from the extension: the 5 V only need to be interrupted in the cable. The extension must be **USB 3.0** (9 contacts, usually blue inside): a USB 2.0 extension lacks the wires the link uses. **Tested: 30 cm cable + about 2 metre extension, a full race with no problems.**

How to remove the 5 V from the cable (any one of these is enough)

- 1** **Cut the red wire (VBUS)** inside the cable, if you can open it.
- 2** **Break or remove pin 1** inside **one** of the two USB-A plugs.
- 3** Use a **small pass-through adapter** that has no pin 1.

Even with the extension, the 5 V pin must be interrupted along the whole cable chain. Always do the two multimeter tests on the next page before connecting.

2. The two multimeter tests (mandatory)

Rules: the 5 V pin must not carry 5 V. No pin may exceed 3.3 V. If a reading is wrong: DO NOT connect the machines.

Multimeter always on **DC volts**, 20 V range. Never touch two neighbouring contacts with the same probe: that is a short circuit. Repeat the tests **every time** you change a cable, an extension, a cartridge or a machine.

Test A: no 5 V through the cable chain

- 1 Assemble the full chain you will use: cable + extension.
- 2 Plug **only one end** of the chain into a **USB charger** (or power bank). The other end stays free.
- 3 The free end has **4 long contacts** and, further back, 5 short ones. Measure **every pair of the 4 long contacts** (6 pairs): they must always read **0 V**.
- 4 If you read about 5 V, the 5 V pin is not interrupted: fix the cable and repeat the test. When finished, **unplug from the charger**.

Test B: on each machine no pin above 3.3 V

- 1 Machine **off**. Connect the cable chain **to this machine only**; the other end stays free. On a Multisystem2 the cartridge must already have the right switch settings (step 4).
- 2 Power the machine on.
- 3 On the free end put the **black probe** on **ground** (pin 4, one of the two outer contacts of the long row). With the **red probe** touch **all the other 8 contacts**, one at a time.
- 4 Every reading must be **between 0 and 3.3 V**. Pin 1 (5 V) must read **0 V**. If you read negative values, the black probe is not on ground: move it to the other outer contact.
- 5 Power off and **repeat on the other machine**. Connect the two machines to each other only with both off.

Contact	What we measured (SuperStation One and Multisystem2)
Pin 1 (5 V, VBUS)	0 V
Pin 4 (ground)	0 V
Pins 2, 3, 5, 6, 7, 8, 9 (signals)	about 3.2 V (correct: not above 3.3 V)

3. What you need

What	Notes
2 MiSTer machines	MiSTer with I/O board, SuperStation One (with the SuperDock, SNAC Bypass switch ON) or Multisystem2 Analogue with Heber's SNAC Classic cartridge (step 4).
1 USB 3.0 A-A cable	With the 5 V pin interrupted (step 1).
1 USB 3.0 extension	Optional, male-to-female; tested at about 2 metres.
A multimeter	For the two tests in step 2.
OutRunners ROMs	Not included. Use the same set that already works with the original MRA.
The files	outrunners.rbf (the core, from the GitHub Releases page) and OutRunners (US) Net Link.mra (from the repository's releases/ folder).

Tested combinations: SuperStation One + MiSTer (link stable for over 30 minutes); Multisystem2 + SuperStation One with an about 2 metre extension (full race, both cars on track).

4. Multisystem2: the SNAC Classic cartridge

The Multisystem2 has no blue USB port: you need Heber's **SNAC Classic Cartridge**, which works only on the **Multisystem2 Analogue** (not the Digital). The cartridge has 3 switches. **From the factory they are all on the left (5 V): that is NOT right for the link.**

Switch	Position	Label on the board	Why
SW1	RIGHT	+3V3 (Signal Lvl)	3.3 V signals. On the left they would be 5 V and reach the other machine.
SW2	RIGHT	+3V3 (Power Lvl)	Lowest supply. With pin 1 interrupted it does not reach the other machine.
SW3	LEFT	IO6 (SNAC)	Pin 9 stays a signal. On the right it would become a 3.3 V supply.

- 1** With the Multisystem2 **off**, set the switches as in the table.
- 2** Insert the cartridge and connect the cable to its blue socket.
- 3** Do **Test B** (step 2) before connecting the other machine.

5. Installing the core on both machines

Same steps and same files on **both**.

- 1 Power the machine off and take out the SD card (or connect over the network).
- 2 Copy **outrunners.rbf** into the **_Arcade/cores/** folder.
- 3 Copy **OutRunners (US) Net Link.mra** into the **_Arcade/** folder.
- 4 Check that the ROMs are in place (usually **games/mame/**).
- 5 **Important:** in **_Arcade/cores/** there must be **only one file starting with «outrunners»**. Move old copies elsewhere.

Where to get the files: outrunners.rbf on the repository's **Releases** page; OutRunners (US) Net Link.mra in the **releases/** folder. That folder also holds the **original core and MRAs without the link**: do not use them to play together.

Checksum (md5) of outrunners.rbf: **5da6d67d9c9b9594b859f3c413dfc1b5** (4592880 bytes). The right core shows the «Comm RAM Clear» and «Comm Hi Byte» entries in the OSD.

6. Connecting the two machines

- 1 Do the two multimeter tests first (step 2).
- 2 **Power off both machines.**
- 3 Connect the cable chain (cable + extension) to the **USER port** of each (on the Multisystem2: the cartridge's blue socket). Do not use the normal USB ports.
- 4 Power both machines back on.

7. OSD settings (on both machines)

Start **OutRunners (US) Net Link** on both, open the OSD (F12 or the mapped button) and set:

OSD entry	Value	Why
Cabinet Link	Network	Turns the link on.
Link Baud	250k	Speed. Same on both . Also tested with the 2 m extension.
Link Test	Off	Link test (step 11): off while playing.
Comm RAM Clear	Once	Default value: leave it.
Comm Hi Byte	FF	Default value: leave it.
Link Role / Cabinet ID	as they are	Ignored by the core: the game decides (step 8).

8. Settings inside the game (Service / Test menu)

The game stores them in its own memory (EEPROM): do them **once** on each machine. If you rename the .mra file they may start again from zero.

- 1 Enter the **Test menu** (OSD «Service Mode» option or the Test button).
- 2 Choose **NETWORK ASSIGNMENTS** (Service button picks the row, Test button changes it).
- 3 Set **COMMUNICATION = NETWORK**, then **PRIVILEGE MODE** and **CABINET ID#** as in the table.
- 4 Choose **EXIT** and leave the Test menu.

Game entry	Machine 1 (Master)	Machine 2 (Slave)
COMMUNICATION	NETWORK	NETWORK
PRIVILEGE MODE	MASTER	SLAVE
CABINET ID#	1	2

One machine Master and the other Slave. Do not use ISDN/Relay mode: it is not supported.

9. Start and play

- 1 **Reset from the OSD on both machines**, a few seconds apart.
- 2 **NETWORK CHECK** appears. If all goes well: **COMMUNICATION SUCCESS**, with THIS MACHINE ID IS 1 (Master) or 2 (Slave).
- 3 Start the race: each player sees the other's car.

10. The LED that tells what is going on

With **Link Test = Off** the disk (HDD) LED flashes as many times as the stage reached, then pauses, then starts again. On the SuperStation the LED may not be visible: watch the screen.

Flashes	Meaning
0 (off)	The game has not turned the network board on (step 8).
1 - 4	The game is doing its initial check.
5	Valid role: messages start going to the other machine.
6	A message arrived from the other machine.
7	Link up.
Fast flicker	The link was up and stopped for more than 3 seconds.

11. Link test (Link Test)

- 1 On **both**: **Cabinet Link = Network**, **Link Test = On**, Link Baud 250k.
- 2 The LED must stay **steadily on**: messages are arriving from the other machine.
- 3 Unplug the cable: the LED goes off after about 1 second. Plug it back: it comes on again.
- 4 Set **Link Test = Off** again.

12. If something goes wrong

What you see	What to do
A multimeter reading above 3.3 V, or 5 V on pin 1	Do not connect. Check the cable (pin 1) and, on the Multisystem2, the cartridge switches (step 4).
NETWORK ERROR / CONNECTION ERROR	Check you have the right core («Comm RAM Clear» and «Comm Hi Byte» entries), only one outrunners file in the cores folder, and both machines in Network (OSD and game).
LED stuck at 5 flashes	The other machine does not answer: cables, Master/Slave, same Link Baud, Reset a few seconds apart.
Works without the extension but not with it	The extension must be USB 3.0 (9 contacts). Run the Link Test; try another extension.
COMMUNICATION TIME OUT	Reset both a few seconds apart.
LED fast flicker	The link dropped: check the cables and Reset.

Limits

- Works only with **two** machines (one Master and one Slave).
- The link uses our own system: **it is not compatible with original cabinets.**
- Test version: it may change.

Base core: Meathax (s32multi). Link code: github.com/bonna79/s32multi-mycore (commit 4fc543f). SNAC Classic cartridge for Multisystem2: Heber Ltd. Cable and pins also tested with Kuba-J's «PSX link cable» (alpha) core.